

Full Stack Development 3

Extend RE Admin with seven enhancements: a dashboard home page, toast notifications, confirmation modals, cookie-based sessions with token validation, header bar refinements, and a transaction tracking feature.

Context

This is Module 8 of the Full-Stack Development Program. You continue in your role as **Junior Developer** at **Genesis Solutions**.

This module is a direct continuation of Module 7. The RE Admin application built last week is functional with basic agent CRUD and a simple login. Rocket Elevators has reviewed the product and returned with a prioritized list of enhancements. This week advances the project from a working prototype into a more polished, production-oriented application.

Client Needs

Rocket Elevators has been using RE Admin to manage agents. During the most recent product review, they identified six areas for improvement:

1. **Better visual feedback**, Users want clear success/error notifications when performing any action. No more silent failures.
2. **Confirmation before destructive actions**, Accidental deletes and unintended updates are a concern. Every significant action must require explicit confirmation.
3. **Persistent sessions**, Users do not want to re-login every time they visit from the same computer. A session lasting 24 hours via cookies is expected.
4. **Dashboard landing page**, A home page with navigation cards for different admin sections, instead of landing directly on the agent list.
5. **Transaction tracking**, A new feature to record and view financial transactions tied to agents, with the last 10 transactions visible and a form to submit new ones.
6. **Header bar refinements**, Move "Create Agent" into the agent management section and display the logged-in user's name in the navbar.

Global Objectives

Through completion of this business simulation, students will demonstrate skills in the following areas:

- React props and state management.
- React Context API for global state.
- Bootstrap UI components with React (React Bootstrap).
- Chart.JS data visualization (research/background).
- Intermediate MongoDB concepts: TTL indexes, aggregate queries.
- Cookies and session token management.

Project Objectives

Extend the existing RE Admin React application by delivering seven features:

1. **Home Page (Card Grid)**, A dashboard landing page with React Bootstrap cards for each admin section.
2. **Toast Notifications**, A global alert system using `AlertContext` and React Bootstrap `<Toast>` components for all user actions.
3. **Confirmation Modals**, A reusable `<ConfirmationModal>` component required before create, update, delete, and transaction submission.
4. **Session Token (Cookie-Based)**, Replace `localStorage` with cookie-based sessions using `react-use-cookie`, UUID token generation, and MongoDB TTL for 24-hour expiration.
5. **Token Validation**, A `useTokenValidation` hook that runs on every page navigation and redirects unauthenticated users to login.
6. **Header Bar Updates**, "Create Agent" moved from navbar to the Agent Management component; logged-in user's fullname displayed in the navbar.
7. **Transaction Feature**, A new Transaction page with a table of the last 10 transactions and a form to submit new ones.

All user-initiated actions must produce toast notifications (green for success, red for error, auto-hiding after 5 seconds). Sessions are cookie-based with a 24-hour TTL enforced by MongoDB. Transaction amounts must accept positive numbers only.

Starting Point

No new repository. Students continue building directly on their existing Module 07 RE Admin project. All seven features are added to the same codebase without any new scaffolding.

Expected Deliverables

- Extended RE Admin with toast notifications, confirmation modals, cookie-based sessions, dashboard, and transactions.
- The `README.md` that clearly explains the project to someone with no prior knowledge of the setup.
- The AI Specification document and its feature specification documents.
- A Postman collection covering all endpoints, exported as `PostmanCollection.json`.
- The `CONCEPTS.md` listing 3 challenging concepts, with a recorded video demonstrating understanding.
- The LeetCode challenge solutions with a recorded video explaining the reasoning, and solution screenshots saved to `./LeetCode-Challenges/<challenge-name>.png`.
- The Technical Demonstration and Code Overview video of the project.
- Any completed and implemented extra miles.

Requirements

The [Requirement Checklist](#) is the line-by-line source of truth for graded evaluation. If there is any conflict between this document and the checklist, the checklist takes precedence.

The checklist evaluates across six categories: project setup, AI-native specifications, technical interview, deadline compliance, extra miles, and professional skills.

Constraints

- **UI Library:** Must install and use **React Bootstrap** components (`npm install react-bootstrap bootstrap`).
- **Session Storage:** Must use **cookies** (via `react-use-cookie`), not `localStorage`.
- **Token Generation:** Must use **UUID** (`uuid` package), not JWT, for session tokens.
- **Session Expiration:** TTL of **24 hours** on the MongoDB Session collection.
- **Notification Duration:** Toasts must auto-hide after **5 seconds**.
- **Transaction Amounts:** Must accept **positive numbers only**.

- **AI Specs required before coding:** Global spec + one feature spec per feature must be written before implementation.
- **Branching Model:** Feature branches (`feature/*`) created from `dev`, merged back to `dev`, then `dev` → `main`. No direct commits to `main`.
- **Grading:** Only the `main` branch will be evaluated.
- **Deadline:** The project must be submitted through the platform by **Friday at 11:59 PM** of the assigned deadline week.
- **Submission Summary:** Must NOT be committed to GitHub, it is submitted separately through the platform.

Professional Expectations

This program evaluates both technical and professional skills. Students are expected to:

- Respond to coaches within **24 hours** when contacted.
- Provide at least **2 progress updates per week**.
- Schedule at least **1 project review per week** before Friday.
- Demonstrate autonomy, initiative, professionalism, and attention to detail.

Failure to meet communication and progress requirements may result in a failed module.

References

- The **Platform Slides** provide teaching guidance, concepts, and setup instructions.
- The **Business Document** is the client project brief, it tells the developer what to build, for whom, and to what standard.
- The [Requirement Checklist](#) for precise information of the requirements to implement.